EXP no:- 11
1/3

Determination of Z Parameters of Two Port network

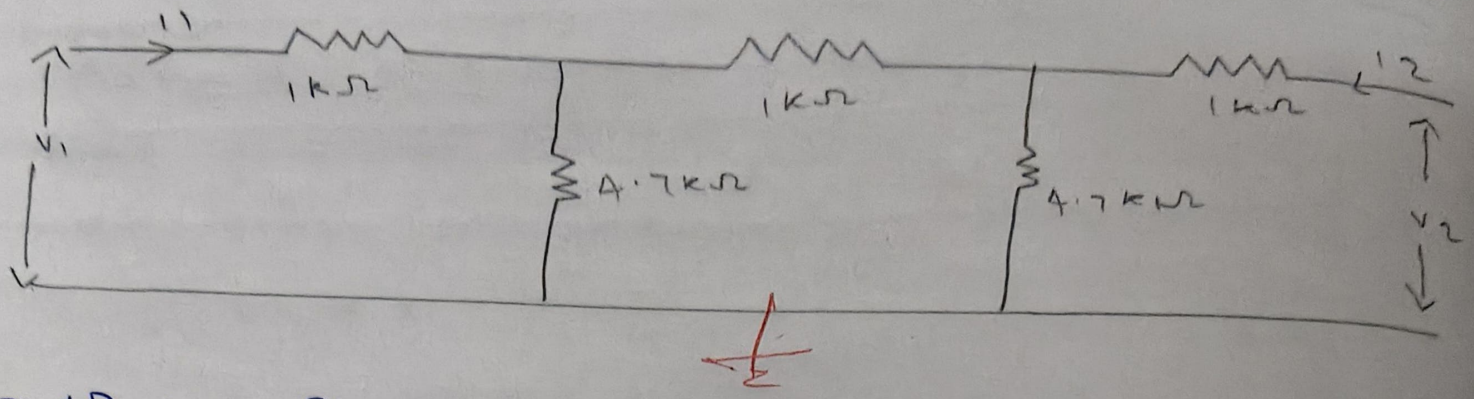
AIM :-

To calculate and verify Z Parameters of a two Port network using multisim simulator.

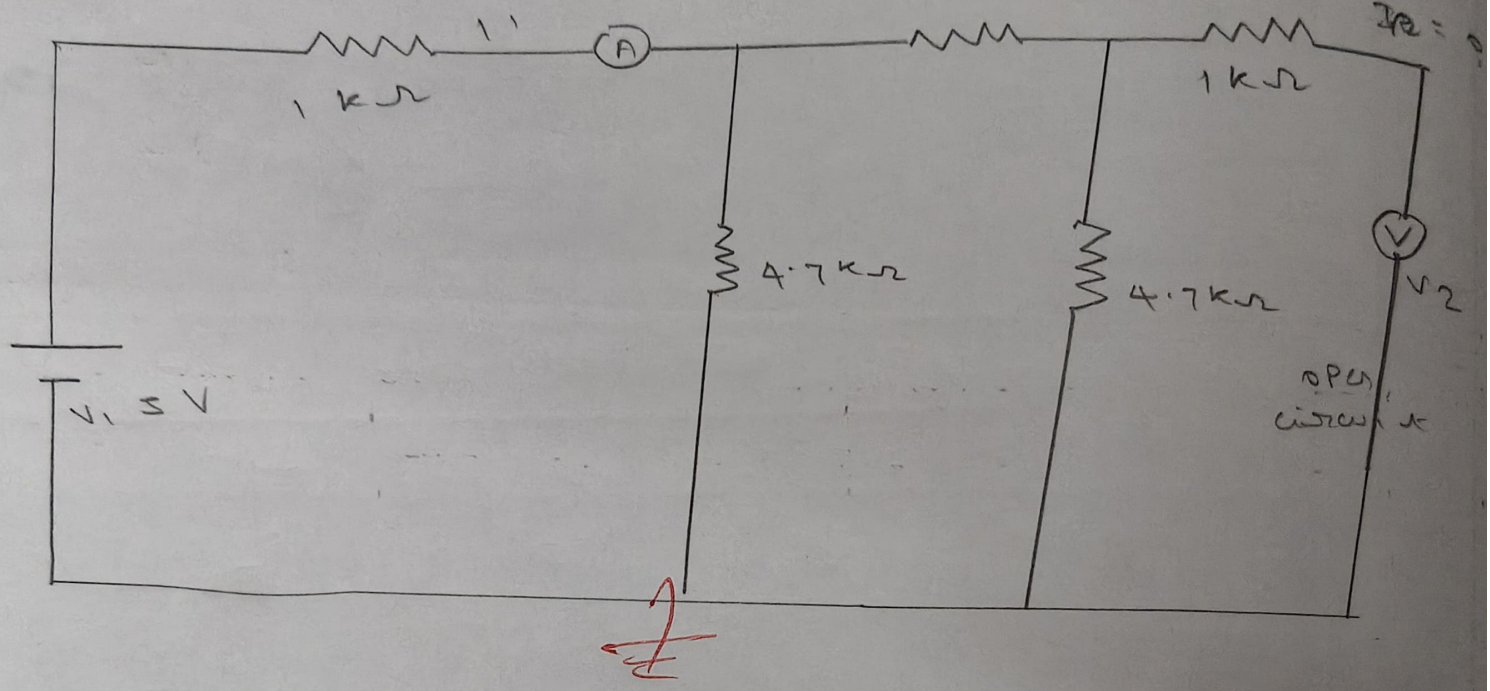
Apparatus required :-

S.NO	Components	Specification	Quantity
1)	Power supply	—	1
2)	multimeter	—	1
3)	Resistors	10kΩ, 10kΩ, 4.7kΩ	—
4)	bread board	—	1
5)	connecting wires	—	—
6)	PC with multisim simulator	—	—

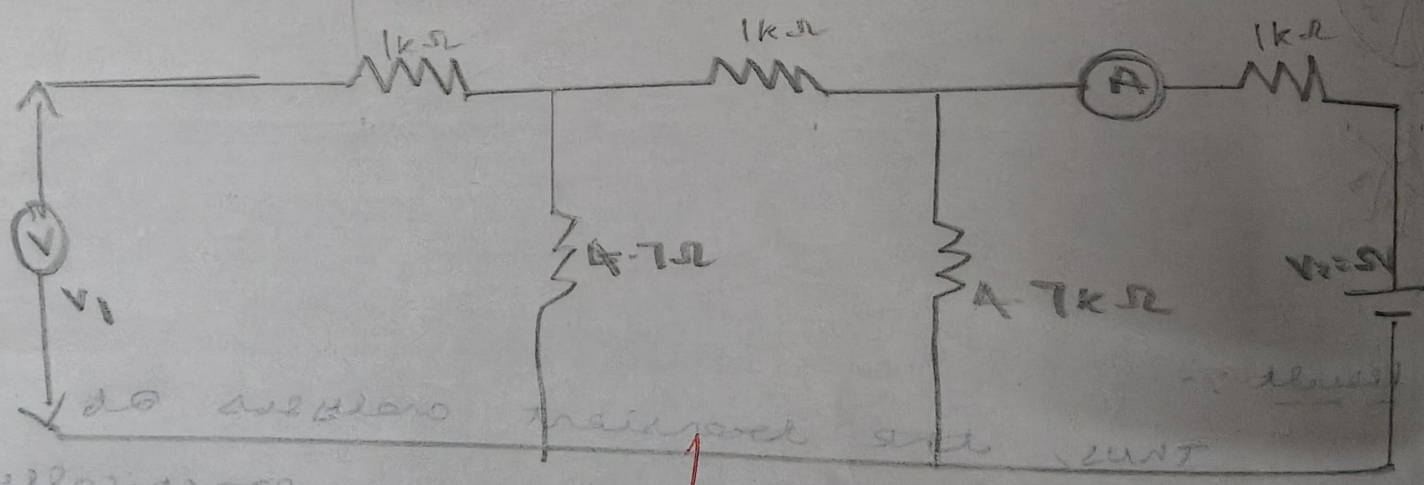
Circuit diagram :-



Output open circuited :-



Input open circuit :-



observation :

S:NO	Output open circuited ($I_2 = 0$)			$Z_{21} = V_2 / I_1$ (Ω)	$Z_{11} = V_1 / I_1$ (Ω)
	V_1	V_2	I_1		
1)	5	2	2	$Z_{21} = \frac{2}{2mA} = 1k\Omega$	$Z_{11} = \frac{5}{2mA} = 2.5k\Omega$
2)	6	3	3	$Z_{21} = \frac{3}{3A} = 1k\Omega$	$Z_{11} = \frac{6}{3mA} = 3k\Omega$

S:NO	Input open circuited ($I_1 = 0$)			$Z_{12} = V_1 / I_2$ (Ω)	$Z_{22} = V_2 / I_2$ (Ω)
	V_2	V_1	I_2		
1)	5	2	2	$Z_{12} = 5/2 = 2.5k\Omega$	$Z_{22} = \frac{5}{2} = 2.5k\Omega$
2)	6	3	3	$Z_{12} = \frac{3}{3} = 1k\Omega$	$Z_{22} = \frac{6}{3} = 2k\Omega$

Op verified

PROCEDURE:

a) correct the circuit as shown in fig and switch 'ON' the experimental board.

b) First open the o/p terminal and supply 5 V to I/P terminal. Measure o/p voltage and I/P current.

c) secondly, open I/P terminal and supply 5V to o/p terminal. Measure I/P voltage and o/p current using multi-meter.

d) calculate the values of z-parameters using equation (1) and (2)

e) switch 'OFF' the supply after taking threedings.

Theortical verification :

$$Z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0}$$

$$= \frac{5}{2mA} = 2.5k\Omega$$

$$Z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0}$$

$$= \frac{2}{2mA} = 1k\Omega$$

$$Z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0}$$

$$= \frac{2}{2mA} = 1k\Omega$$

$$Z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0}$$

$$= \frac{5}{mA} = 2.5k\Omega$$

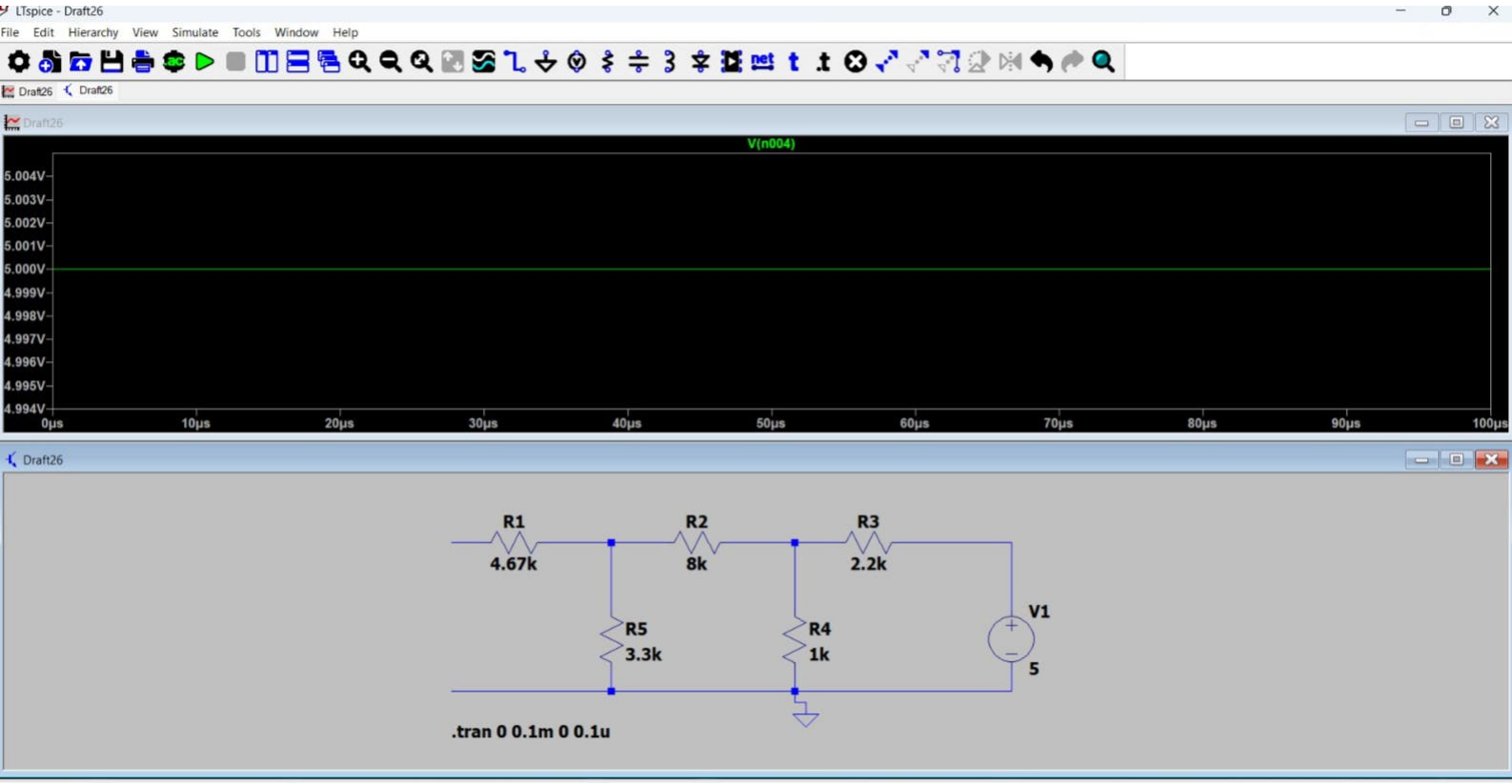
$$Z = \begin{bmatrix} 2.5k\Omega & 1k\Omega \\ 1k\Omega & 2.5k\Omega \end{bmatrix}$$

Marks split - up :-

S.No	Description	Marks allotted	Marks obtained
1)	Experimental / setup / correction	20 marks	20
2)	observation / readings	20 marks	10
3)	calculations / result	20 marks	20
4)	viva	20 marks	20
5)	Record	20 marks	20
			90

Result :-

The z- parameters of the two port network has been calculated and verified by using multisim simulator.


LTspice - Draft26
File Edit Hierarchy View Simulate Tools Window Help
Draft26
V(n004)
5.004V
5.003V
5.002V
5.001V
5.000V
4.999V
4.998V
4.997V
4.996V
4.995V
4.994V
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
R1
4.67k
R2
8k
R3
2.2k
R5
3.3k
R4
1k
V1
5
.tran 0 0.1m 0 0.1u